SAR LAB 7 – LANDSLIDE CASE STUDY

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Space Technologies (Downstream)

1. Introduction

The objective of this study is to perform a comprehensive remote sensing analysis of the catastrophic landslide that occurred in Xinmo Village, Maoxian County, Sichuan Province, China, on June 24, 2017. This study replicates the scientific workflow by combining InSAR Time Series Analysis (SAR) using the COMET-LiCSAR dataset and Optical Remote Sensing Analysis (Sentinel-2) to map land cover changes and vegetation loss.

2. InSAR Time Series Analysis

2.1 Methodology

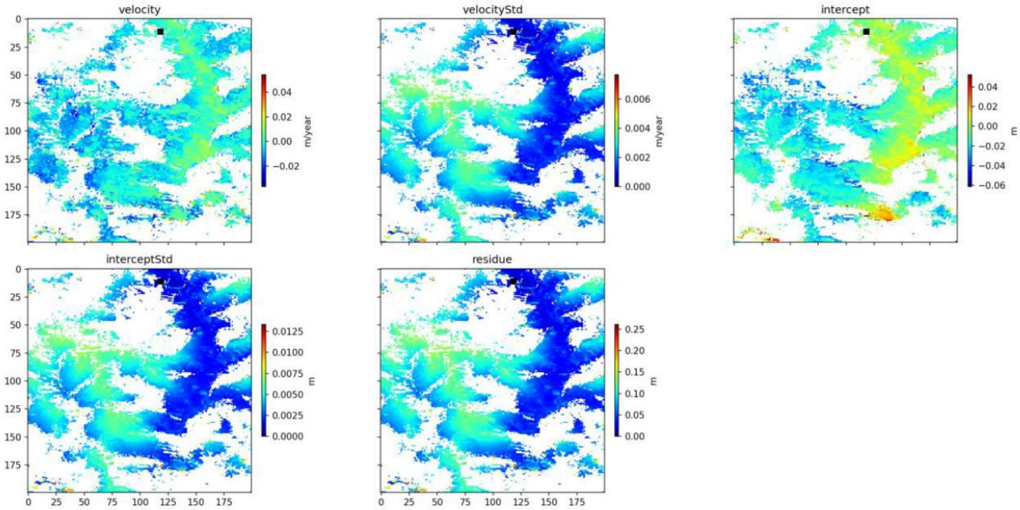
The deformation history preceding the landslide was reconstructed using the Small Baseline Subset (SBAS) InSAR technique via MintPy. The dataset covers the period from late 2014 to June 2017.

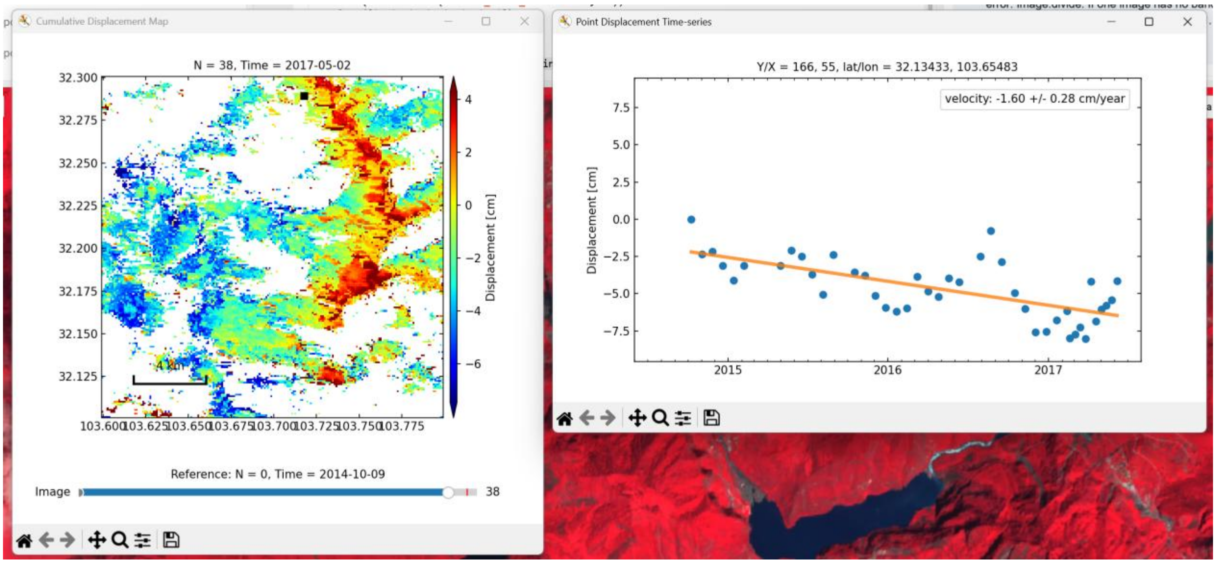
To mitigate spatial noise and atmospheric phase screens (APS), a stable reference point was selected outside of the active landslide body. The chosen reference coordinates are lat/lon = 32.13433, 103.65483. Selecting a stable reference point near, but outside, the landslide is critical as it serves as a zero-deformation baseline, allowing for accurate relative displacement measurements of the moving slope.

2.2 Results

The generated InSAR velocity map clearly outlines the active deformation zone on the high-altitude slope. The pre-event displacement rates show significant motion concentrated in the upper and middle parts of the failure body.

The displacement history of the selected highly active pixel (shown in the time-series plot) reveals a steady, linear downward trend from 2015 leading up to the final failure in June 2017. The slope exhibited a constant pre-failure displacement rate of -1.98 ± 0.20 cm/year. This progressive deformation indicates a slow, gravitational slope instability that ultimately culminated in the catastrophic failure.





3. Coherence loss

InSAR coherence (cross-correlation between two SAR images) is highly sensitive to changes in surface geometry. When a landslide occurs, the local topography is completely reorganized: vegetation is stripped, and soil/rock is displaced. This causes a total loss of coherence (decorrelation). Coherence loss is an excellent and highly sensitive tool for mapping the deposition zone (runout area) of a landslide. By comparing a pre-event/post-event interferogram, the landslide scar appears as a distinct, low-coherence (dark) shape cutting through the otherwise coherent surrounding terrain.

Sentinel-1 GRD (Ground Range Detected) backscatter in the co-polarized VV channel is sensitive to surface roughness and soil moisture. Before the landslide, the slope was covered by a dense forest, which causes significant volumetric scattering (lower VV, higher VH). After the landslide, the bare rock and rough debris field act as surface scatterers, significantly altering the VV backscatter intensity. Compared to coherence loss, VV change is highly useful for identifying changes in surface material (e.g., forest to bare rock), but it can be noisy due to changes in soil moisture and local slope geometry. Therefore, coherence loss remains superior for mapping the precise spatial extent (boundaries) of the landslide, while VV change is better for analyzing physical surface changes.

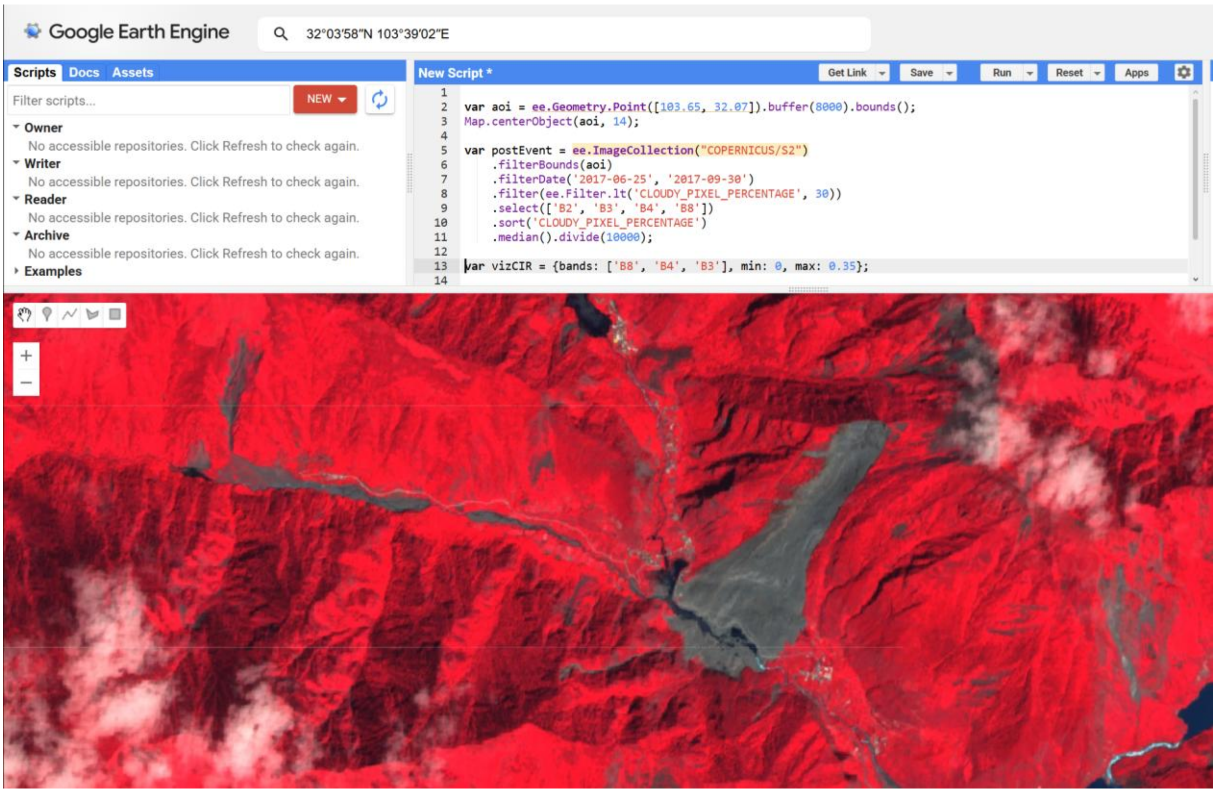
4. Optical analysis

4.1 Methodology

To map the landslide footprint and quantify vegetation loss, cloud-free Sentinel-2 L1C image from June 25, 2017 – September 30, 2017 was used. To ensure high-quality results in this high-altitude region (elevations > 3,000 m), the search window was constrained to late spring and summer to avoid snow cover. A cloud filter of 30% was applied. The imagery was visualized using a False-Color Infrared (CIR) composite (Bands: B8 (NIR), B4 (Red), B3 (Green)). In this composite, healthy vegetation appears in vibrant red due to high Near-Infrared reflectance, while bare soil and rock appear in grey/cyan.

4.2 Results

A massive, distinct grey/cyan scar is visible, cutting through the red vegetation background. This scar represents the path of the landslide.



5. Conclusions

This laboratory exercise successfully demonstrated the synergy of combining SAR and Optical datasets. InSAR (MintPy) acted as an early-warning diagnostic tool, capturing a steady pre-failure displacement rate of ~1.98 cm/year over two years prior to the disaster. Optical CIR (Sentinel-2) provided an instant, high-contrast post-event mapping tool that clearly delineated the landslide's spatial footprint and highlighted the severe ecological impact (vegetation loss).